

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
1st Semester of M.Sc.(Mathematics)Theory Examination November 2018
MA 714:Ordinary Differential Equations

Date: 29/11/2018,Thursday**Time:** 10:00 a.m. to 10:45 a.m. **Maximum Marks:** 20

Multiple Choice Questions

Instructions:

- (1) Tick the correct answer only in this sheet.
 - (2) You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
 - (3) Each question carries one mark.
 - (4) In this question paper, $\mathbb{R}$ denotes the set of all real numbers and $\mathbb{Z}$ denotes the set of all integers.
-

Q.1. Choose the correct answer for the following questions. [20]

1. The complementary function of $y'' - 6y' + 9y = 2xe^{\sin x}$ is _____, where A and B are arbitrary constants.

a. $(A + Bx)e^{3x}$ c. $Ae^{3x} + Be^{-3x}$
b. $(A + Bx)e^{-3x}$ d. $A \cos 3x + B \sin 3x$.

2. The particular integral of $y'' - y = e^{5x}$ is _____.

a. $\frac{e^{5x}}{25}$ b. $-\frac{e^{5x}}{25}$ c. $\frac{e^{5x}}{24}$ d. $-\frac{e^{5x}}{24}$.

3. The set of linearly independent solutions of the differential equation $\frac{d^4 y}{dx^4} - \frac{d^2 y}{dx^2} = 0$ is _____.

a. $\{1, x, e^x, e^{-x}\}$ c. $\{1, x, e^x, xe^x\}$
b. $\{1, x, e^{-x}, xe^{-x}\}$ d. $\{1, x, e^x, xe^{-x}\}$

4. What is the type of the differential equation $\frac{d^2 y}{dx^2} + y^2 \frac{dy}{dx} - x^2 y = \sin x$?

a. linear and homogenous
b. non-Linear and non-homogenous
c. linear and non-homogenous
d. non-Linear and homogenous

5. The differential equation whose linearly independent solutions are $\cos 7x$ and $\sin 7x$ is _____.

a. $y'' - 49y = 0$ c. $y'' - 7y = 0$
b. $y'' + 49y = 0$ d. $y'' + 7y = 0$.

6. If y_1 is a solution of the differential equation $y'' + P(x)y' + Q(x)y = 0$, then other solution of the equation is $y_2 = vy_1$, where $v =$ _____.

a. $\int \frac{1}{y_1^2} e^{-\int P(x) dx} dx$ c. $\int \frac{1}{y_1} e^{-\int P(x) dx} dx$
b. $\int \frac{1}{y_1^2} e^{\int P(x) dx} dx$ d. $\int \frac{1}{y_1} e^{\int P(x) dx} dx$.

7. The maximum number of linearly independent solutions of the differential equation $\frac{d^4 y}{dx^4} = 0$, with the condition $y(0) = 1$ is _____.

a. 1 b. 2 c. 3 d. 4.

8. For the differential equation $x^2(1-x)y'' + xy' + y = 0$, which of the following is true?

a. $x = 1$ is an ordinary point
b. $x = 1$ is a regular singular point
c. $x = 0$ is an irregular singular point
d. $x = 0$ is an ordinary point

9. Which of the following points is the irregular singular point of the differential equation $x^5(x-8)(x-9)y'' - x(x-8)y' + 13(x-9)y = 0$?
- a. 0 b. 8 c. 9 d. 13
10. Let ξ be a regular singular point of the differential equation $y'' + P(x)y' + Q(x)y = 0$. Then this equation has at least one non-trivial solution of the form _____ valid in a neighborhood of ξ , where $a_0 \neq 0$.
- a. $\sum_{n=0}^{\infty} a_n(x-\xi)^n$ c. $(x-\xi)^m \sum_{n=0}^{\infty} a_n x^n$
b. $(x-\xi)^m \sum_{n=0}^{\infty} a_n(x-\xi)^n$ d. $(x-\xi)^m \sum_{n=0}^{\infty} a_n x^{n+m}$.
11. Let m_1 and m_2 be two roots of indicial equation of differential equation $y'' + P(x)y' + Q(x)y = 0$ and the equation has a solution $x^{m_1} \sum_{n=0}^{\infty} a_n x^n$ ($a_0 \neq 0$). If m_1 and m_2 not differ by integer, then the equation has an another independent solution _____.
- a. $x^{m_2} \sum_{n=0}^{\infty} a_n x^{n+m_2}$ c. $x^{m_1} \sum_{n=0}^{\infty} a_n x^{n+m_2}$
b. $x^{m_2} \sum_{n=0}^{\infty} a_n x^n$ d. none of these.
12. The particular solution of differential equation $x(1-x)y'' + [c - (a+b+1)x]y' - aby = 0$, where a, b and c are constants is called_____.
- a. Bessel function c. Hermite function
b. Legendre function d. hypergeometric function.
13. The radius of convergence of the series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ is_____.
- a. ∞ b. 5 c. 2 d. 1
14. Which of the following is the auxiliary equation for the system of the differential equations $\frac{dx}{dt} = x + y$; $\frac{dy}{dt} = 4x - 2y$?
- a. $m^2 - m + 6 = 0$ c. $m^2 + m - 6 = 0$
b. $m^2 - m - 6 = 0$ d. $m^2 + m + 6 = 0$.

15. What is the Wronskian of two solutions $x = x_1(t), y = y_1(t)$ and $x = x_2(t), y = y_2(t)$ of the system of differential equations $\frac{dx}{dt} = a_1(t)x + b_1(t)y$; $\frac{dy}{dt} = a_2(t)x + b_2(t)y$?

a. $\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$

b. $\begin{vmatrix} y_1 & x_1 \\ y_2 & x_2 \end{vmatrix}$

c. $\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}$

d. $\begin{vmatrix} y_1 & x_1 \\ x_2 & y_2 \end{vmatrix}$

16. The solution of Volterra's predator-prey population equations is a family of _____.

a. circles

b. parabolas

c. hyperbolas

d. ellipses.

17. Let m_1 and m_2 be two complex numbers with non zero real parts. If m_1 and m_2 are roots of auxiliary equation of the system of differential equations $\frac{dx}{dt} = a_1(t)x + b_1(t)y$; $\frac{dy}{dt} = a_2(t)x + b_2(t)y$. Then the critical point $(0, 0)$ of the system is _____.

a. node

b. spiral

c. saddle point

d. none of these

18. Which of the following systems of differential equations is not linear?

a. $\frac{dx}{dt} = x \log t + y$; $\frac{dy}{dt} = x + y \cos t$

b. $\frac{dx}{dt} = y$; $\frac{dy}{dt} = x$

c. $\frac{dx}{dt} = t^3x + y^2$; $\frac{dy}{dt} = 5x^3 - y \sin t$

d. $\frac{dx}{dt} = \sec tx - ty$; $\frac{dy}{dt} = -5x - ty$.

19. The initial value problem $y' = f(x, y), y(x_0)$, where $f(x, y)$ is an arbitrary function defined and continuous in some neighborhood of the point (x_0, y_0) is equivalent to the integral equation _____.

a. $y(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt$

b. $y(x) = y_0 \int_{x_0}^x f(t, y(t)) dt$

c. $y(x) = x_0 + \int_{y_0}^x f(t, y(t)) dt$

d. none of these

20. A function $f(x, y)$ is said to satisfy Lipschitz condition in the variable x if _____.

a. $|f(x, y_1) - f(x, y_2)| \leq K|y_1 - y_2|$

b. $|f(x, y_1) - f(x, y_2)| \leq K|x_1 - x_2|$

c. $|f(x_1, y) - f(x_2, y)| \leq K|x_1 - x_2|$

d. $|f(x_1, y) - f(x_2, y)| \leq K|y_1 - y_2|$,

for some $K > 0$.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
1st Semester of M.Sc.(Mathematics)Theory Examination November 2018
MA 714 Ordinary Differential Equations

Date: 29/11/2018, Thursday **Time:** 10:45 a.m. to 01:00 p.m. **Maximum Marks:** 50

Instructions:

- (5) Section I and II must be written in separate answer sheets
 - (6) You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
 - (7) Figures to the right indicate marks.
-

SECTION-I

Q-2(a) Attempt any two of the following: [08]

- (i) Using method of variation of parameters, solve the differential equation
$$y'' - 6y' + 9y = \frac{e^{3x}}{x^2}.$$
- (ii) Solve the differential equation $y'' + 4y = 4 \sin 2x$ using method of undetermined coefficients.
- (iii) Find the general solution of the system:

$$\frac{dx}{dt} = x + y; \quad \frac{dy}{dt} = 4x - 2y.$$

(b) Attempt any four of the following: [12]

- (i) Locate and classify the singular points of the differential equation
$$x^5(x-1)(x-5)^2y'' - 6x(x-1)(x-5)y' + 7y = 0.$$
- (ii) For hypergeometric function $F(a, b, c, x)$, prove that
$$x \left[\lim_{a \rightarrow \infty} F(a, a, \frac{3}{2}, -\frac{x^2}{4a^2}) \right] = \sin x.$$
- (iii) If $y_1(x)$ and $y_2(x)$ are two solutions of differential equation $y'' + P(x)y' + Q(x)y = 0$ on $[a, b]$, then show that their Wronskian $W(y_1, y_2)$ is either identically zero or never zero.

- (iv) If e^x is a solution of the differential equation $xy'' - (2x + 1)y' + (x + 1)y = 0$, then find its general solution.
- (v) Determine the nature of the critical point $(0, 0)$ for the following linear autonomous system:
- $$\frac{dx}{dt} = 4x - 3y; \quad \frac{dy}{dt} = 8x - 6y.$$
- (vi) Using Picard's method, solve $\frac{dy}{dx} + xy = 0$ with $y(0) = 1$ upto third approximation.

SECTION II

Q-3(a) **Attempt any three of the following**

[18]

- (i) Solve the differential equation $(3x + 2)^2 y'' + 3(3x + 2)y' - 36y = 3x^2 + 4x + 1$.
- (ii) Find the particular solution of the non homogenous system
- $$\frac{dx}{dt} = x + y - 5t + 2; \quad \frac{dy}{dt} = 4x - 2y - 8t - 8.$$
- (iii) Find the series solution of the differential equation $xy'' + y' - y = 0$.
- (iv) Find the series solution of the Chebyshev differential equation $(1 - x^2)y'' - xy' + p^2y = 0$, where p is constant.
- (v) Find the series solution of the differential equation $2x(1 - x)y'' + (1 - x)y' + 3y = 0$.

(b) **Attempt any two of the following**

[12]

- (i) Let $f(x, y)$ be a continuous function satisfy a Lipschitz condition $|f(x, y_1) - f(x, y_2)| \leq |y_1 - y_2|$ on a strip defined by $a \leq x \leq b$ and $-\infty < y < \infty$. If (x_0, y_0) is any point of the strip, then show that the initial value problem $y' = f(x, y), y(x_0) = y_0$ has unique solution on the interval $a \leq x \leq b$.

- (ii) Define ordinary point of differential equation. Let origin be an ordinary point of the differential equation $y'' + P(x)y' + Q(x)y = 0$, and let a_0 and a_1 be arbitrary constants. Then prove that there exists a unique function $y(x)$ which is a solution of $y'' + P(x)y' + Q(x)y = 0$ and satisfies the initial conditions $y(0) = a_0$ and $y'(0) = a_1$.
- (iii) (a) State and prove principle of superposition for differential equation $y'' + P(x)y' + Q(x)y = 0$.
- (b) Suppose $y_1(x)$ and $y_2(x)$ are two solutions of the differential equation $y''(x) + P(x)y'(x) + Q(x)y(x) = 0$ on $[a, b]$. Show that they are linearly dependent on $[a, b]$ if and only if their Wronskian $W(y_1, y_2)$ is identically zero.

* * * * *